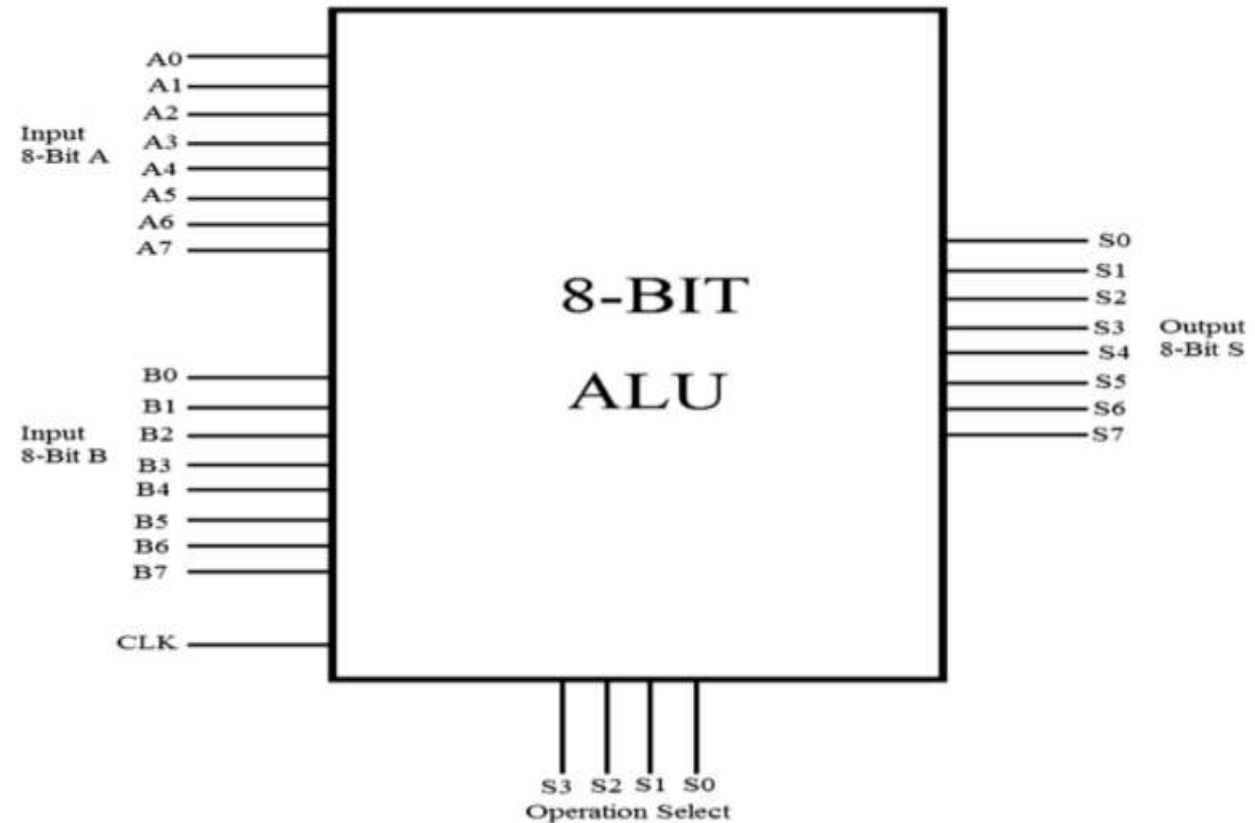


RTL-GDS for an clock based 8 bit ALU

Architecture



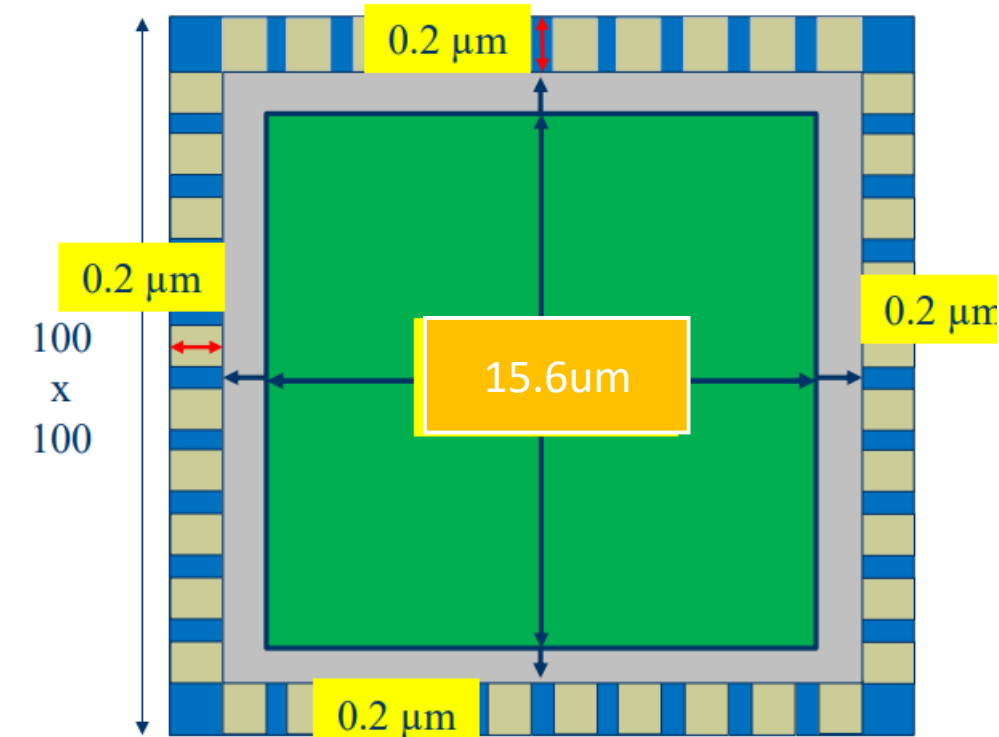
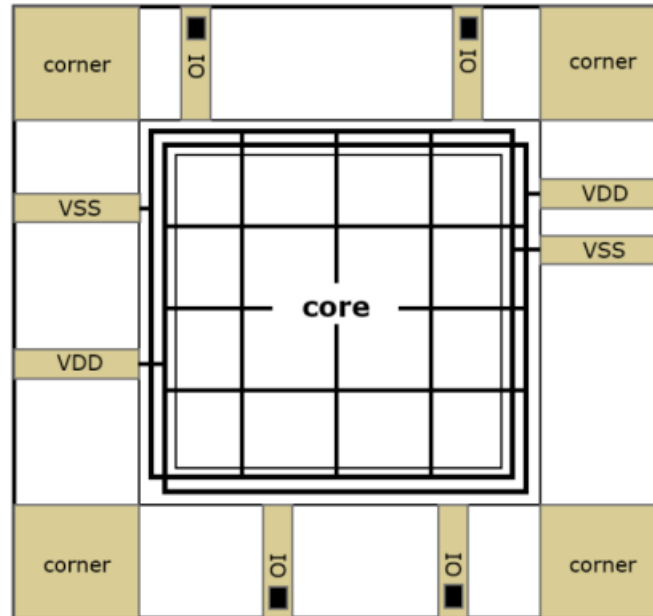
I/O planning

At top view of design we basically have an 8 bit ALU with [7:0]A , [7:0]B , [3:0]sel , 1 clk and 1 rst. So totally input are $8+8+4+1+1=22$ inputs and outputs are [7:0]res so 8 output pins hence . Totally 30 I/O pins . Through our RTL code we will only logically instantiate these 30 pins . But for Area calculations we will take few more 2 more dummy pins hence 32 and for every 8 I/O pin we will take 1VDD and 1VSS pin , hence totally $32+8=40$ pins. Each IO pin take 16mA current , hence the chip should be able to sustain at least $32*16\text{mA}=512\text{mA}$ current.

Proper GDS2 stream out with IO pad instantiation

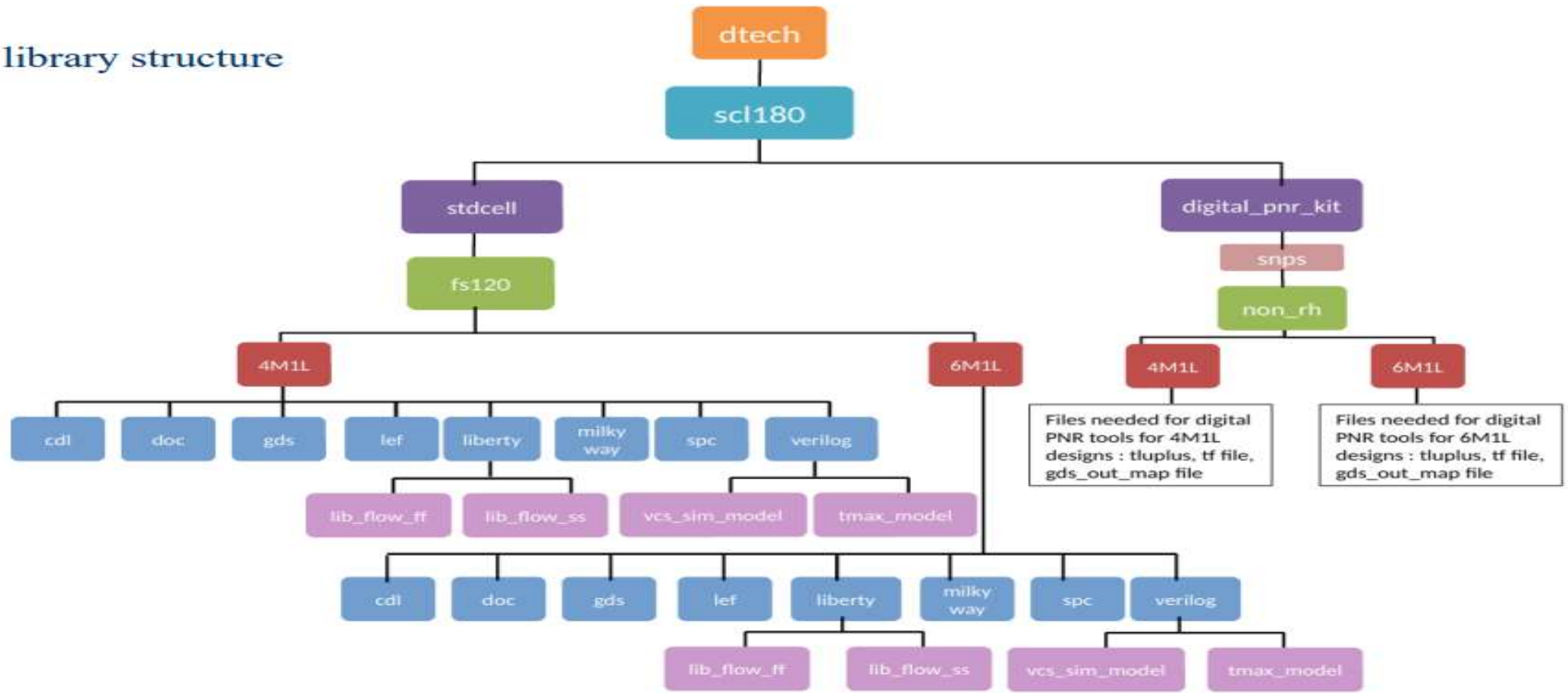
Area Planning

Sr. No.	Die Size (mm)	Package Size (mm)	Pin Count	Package Lead Pitch (mm)	CoB Package Type
1	2.0 x 2.0	4.0 x 4.0	16	0.65	QFN/DIP
2	3.0 x 3.0	5.0 x 5.0	16	0.8	QFN/DIP
3	2.0 x 2.0	4.0 x 4.0	24	0.65	QFN/DIP
4	3.0 x 3.0	5.0 x 5.0	24	0.65	QFN/DIP
5	3.0 x 3.0	5.0 x 5.0	28	0.5	QFN/DIP
6	2.0 x 2.0	5.0 x 5.0	32	0.5	QFN/DIP
7	3.0 x 3.0	5.0 x 5.0	32	0.5	QFN/DIP
8	2.0 x 2.0	7.0 x 7.0	48	0.5	QFN/DIP
9	3.0 x 3.0	7.0 x 7.0	48	0.5	QFN/DIP
10	2.0 x 2.0	9.0 x 9.0	64	0.5	QFN/DIP
11	3.0 x 3.0	9.0 x 9.0	64	0.5	QFN/DIP
12	4.0 x 4.0	7.0 x 7.0	48	0.5	QFN/DIP
13	5.0 x 5.0	7.0 x 7.0	48	0.5	QFN/DIP
14	4.0 x 4.0	9.0 x 9.0	64	0.5	QFN/DIP
15	5.0 x 5.0	9.0 x 9.0	64	0.5	QFN/DIP
16	4.0 x 4.0	10.0 x 10.0	72	0.5	QFN/PGA
17	5.0 x 5.0	10.0 x 10.0	72	0.5	QFN/PGA
18	4.0 x 4.0	12.0 x 12.0	80	0.5	QFN/PGA
19	5.0 x 5.0	12.0 x 12.0	80	0.5	QFN/PGA
20	4.0 x 4.0	12.0 x 12.0	88	0.5	QFN/PGA
21	5.0 x 5.0	12.0 x 12.0	88	0.5	QFN/PGA
22	4.0 x 4.0	14.0 x 14.0	100	0.5	QFN/PGA
23	5.0 x 5.0	14.0 x 14.0	100	0.5	QFN/PGA



Understanding the scl180 library

Standard cell library structure



RTL code(pad instantiation part)

```
pc3c01 pc3c01_1 (.CCLK(alu_clk), .CP(main_clk) );
pc3d01 pc3d01_1 (.PAD(main_clk_pad), .CIN(alu_clk) );
pc3d01_2 (.PAD(main_reset_pad), .CIN(main_reset) );
// A[7:0]
pc3d01 pc3d01_3 (.PAD(A_pad[0]), .CIN(A[0]) );
pc3d01 pc3d01_4 (.PAD(A_pad[1]), .CIN(A[1]) );
pc3d01 pc3d01_5 (.PAD(A_pad[2]), .CIN(A[2]) );
pc3d01 pc3d01_6 (.PAD(A_pad[3]), .CIN(A[3]) );
pc3d01 pc3d01_7 (.PAD(A_pad[4]), .CIN(A[4]) );
pc3d01 pc3d01_8 (.PAD(A_pad[5]), .CIN(A[5]) );
pc3d01 pc3d01_9 (.PAD(A_pad[6]), .CIN(A[6]) );
pc3d01 pc3d01_10 (.PAD(A_pad[7]), .CIN(A[7]) );

// B[7:0]
pc3d01 pc3d01_11 (.PAD(B_pad[0]), .CIN(B[0]) );
pc3d01 pc3d01_12 (.PAD(B_pad[1]), .CIN(B[1]) );
pc3d01 pc3d01_13 (.PAD(B_pad[2]), .CIN(B[2]) );
pc3d01 pc3d01_14 (.PAD(B_pad[3]), .CIN(B[3]) );
pc3d01 pc3d01_15 (.PAD(B_pad[4]), .CIN(B[4]) );
pc3d01 pc3d01_16 (.PAD(B_pad[5]), .CIN(B[5]) );
pc3d01 pc3d01_17 (.PAD(B_pad[6]), .CIN(B[6]) );
pc3d01 pc3d01_18 (.PAD(B_pad[7]), .CIN(B[7]) );
```

```
// sel[3:0]
pc3d01 pc3d01_19 (.PAD(sel_pad[0]), .CIN(sel[0]) );
pc3d01 pc3d01_20 (.PAD(sel_pad[1]), .CIN(sel[1]) );
pc3d01 pc3d01_21 (.PAD(sel_pad[2]), .CIN(sel[2]) );
pc3d01 pc3d01_22 (.PAD(sel_pad[3]), .CIN(sel[3]) );

//=====
// Output Pad Instantiations
//=====
// result[7:0]
pc3o05 pc3o05_1 (.I(result[0]), .PAD(result_pad[0]) );
pc3o05 pc3o05_2 (.I(result[1]), .PAD(result_pad[1]) );
pc3o05 pc3o05_3 (.I(result[2]), .PAD(result_pad[2]) );
pc3o05 pc3o05_4 (.I(result[3]), .PAD(result_pad[3]) );
pc3o05 pc3o05_5 (.I(result[4]), .PAD(result_pad[4]) );
pc3o05 pc3o05_6 (.I(result[5]), .PAD(result_pad[5]) );
pc3o05 pc3o05_7 (.I(result[6]), .PAD(result_pad[6]) );
pc3o05 pc3o05_8 (.I(result[7]), .PAD(result_pad[7]) );
```

Proper GDS2 stream out with IO pad instantiation

Actual Logic

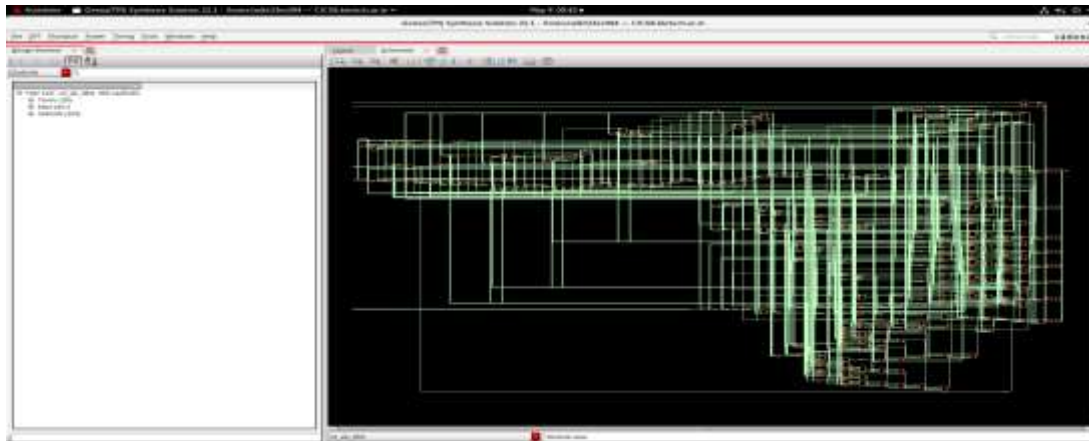
```
always @(*) begin
  case (sel)
    4'b0000: alu_out = A + B;      // Addition
    4'b0001: alu_out = A - B;      // Subtraction
    4'b0010: alu_out = A + 1;      // Increment
    4'b0011: alu_out = A - 1;      // Decrement
    4'b0100: alu_out = A * B;      // Multiplication
    4'b0101: alu_out = (B != 0) ? A / B : 8'b00000000; // Division
    4'b0110: alu_out = A << 1;     // Logical left shift
    4'b0111: alu_out = A >> 1;     // Logical right shift

    4'b1000: alu_out = A & B;      // AND
    4'b1001: alu_out = A | B;      // OR
    4'b1010: alu_out = A ^ B;      // XOR
    4'b1011: alu_out = ~A;         // NOT A
    4'b1100: alu_out = ~(A & B);   // NAND
    4'b1101: alu_out = ~(A | B);   // NOR
    4'b1110: alu_out = ~(A ^ B);   // XNOR
    4'b1111: alu_out = 8'b00000000; // Clear output
    default: alu_out = 8'b00000000;
  endcase
end

// Registered output
always @(posedge main_clk or posedge main_reset) begin
  if (main_reset)
    result <= 8'b00000000;
  else
    result <= alu_out;
end
endmodule
```


Proper GDS2 stream out with IO pad instantiation

Results after synthesis



```
@genus:root: 90> report_area
=====
Generated by:      Genus(TM) Synthesis Solution 20.11-s111_1
Generated on:      Feb 18 2026  05:19:16 pm
Module:           alu_8b_wps
Operating conditions:  ts18c1o150_max (worst_case_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Instance  Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
alu_8b_wps          395  47822.792  254.228   48077.020  70000 (S)
(S) = wireload was automatically selected
@genus:root: 91>
```

```
Path 1: MET (18814 ps) Setup Check with Pin result_reg[0]/CP->D
Group: I2R
Startpoint: (F) B_pad[5]
Clock: (R) vlr_main_clk_i
Endpoint: (R) result_reg[0]/D
Clock: (R) alu_clk

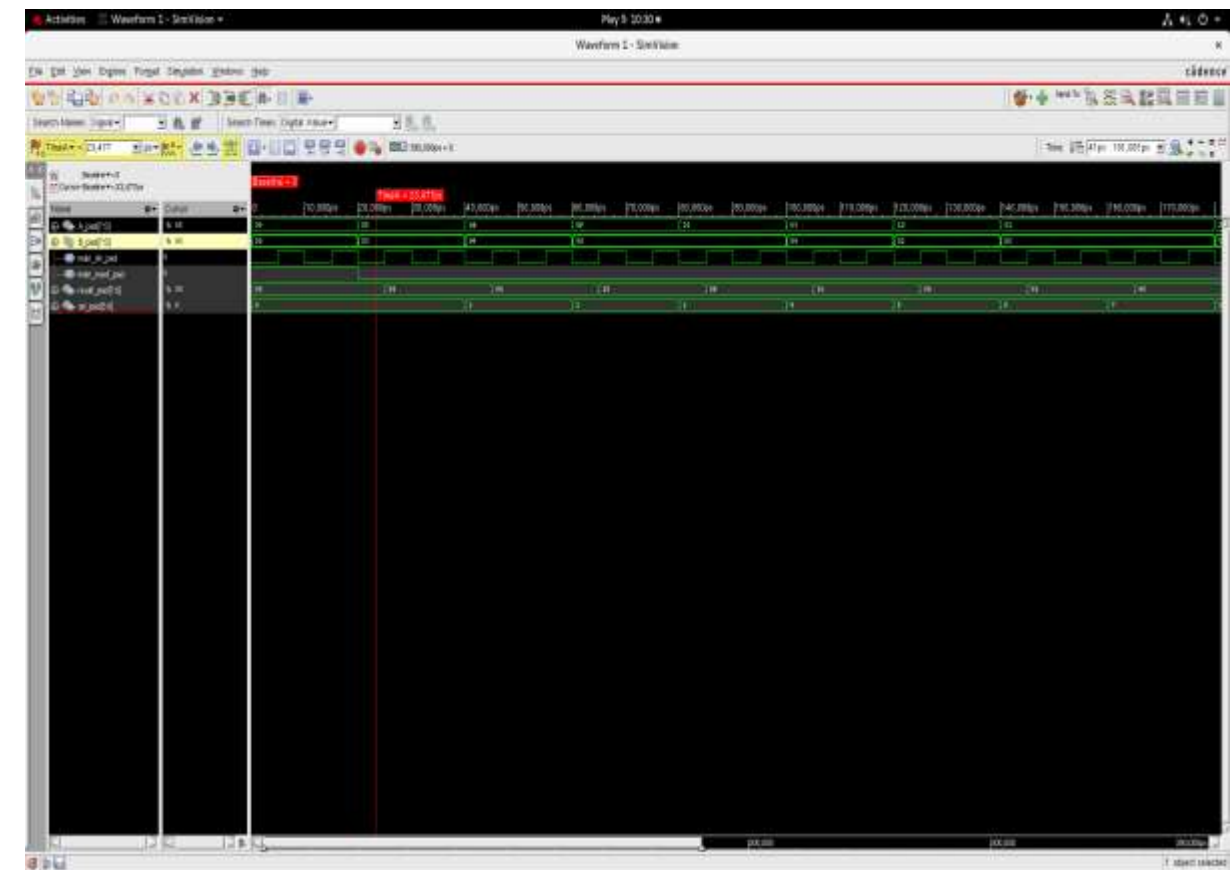
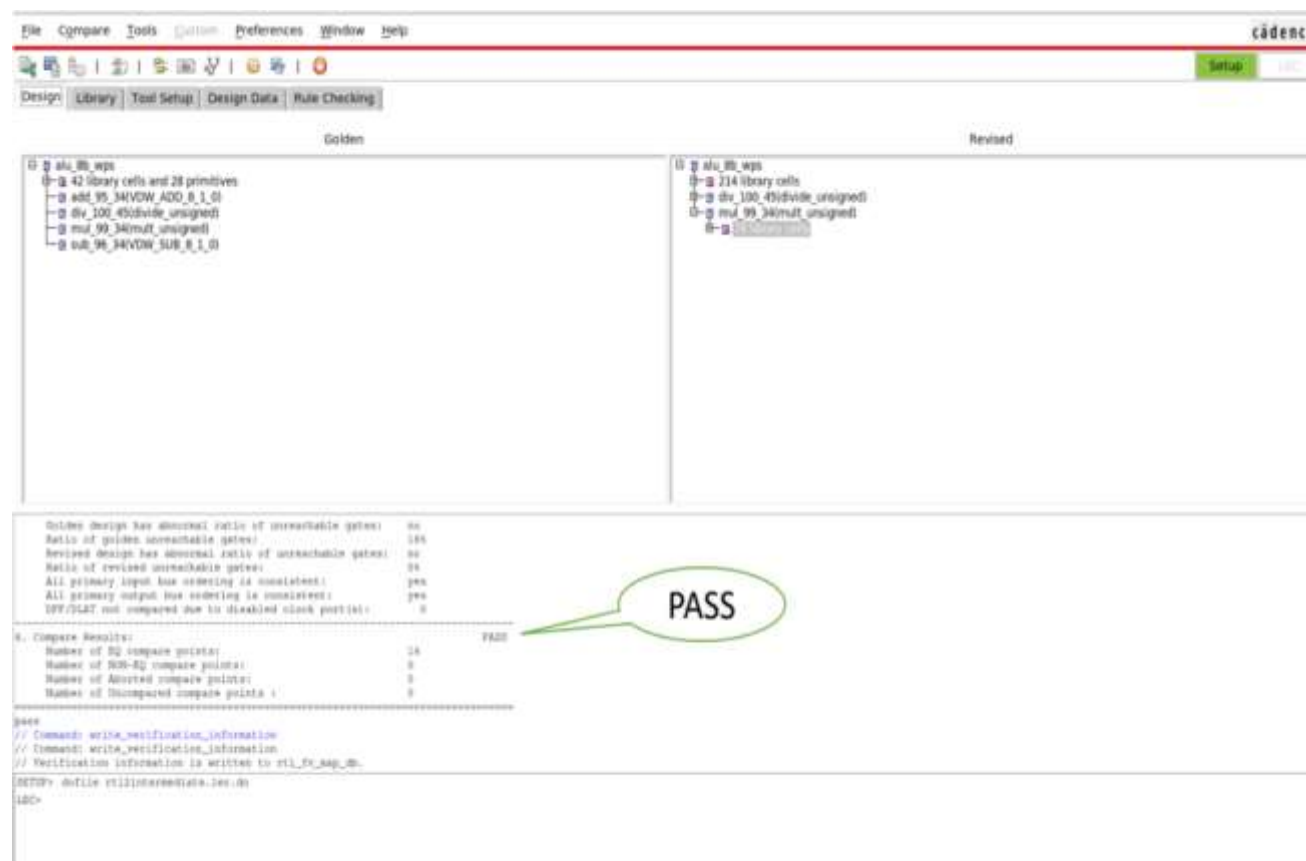
Capture          Launch
Clock Edge:+     0
Drv Adjust:+     0
Src Latency:+    1000
Net Latency:+    0 (I)
Arrival:=        51000

Setup:-          282
Uncertainty:-    1250
Path Adjust:+    -1500
Required Time:=  47968
Launch Clock:-   0
Input Delay:-    7750
Data Path:-      21404
Slack:=          18814
```

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	6.98480e-09	2.18073e-05	2.25426e-05	4.43569e-05	1.43%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	8.85597e-08	4.37538e-05	5.23952e-05	9.62376e-05	3.10%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	1.85703e-04	1.85703e-04	5.98%
pad	1.17229e-05	1.72797e-03	1.04166e-03	2.78135e-03	89.50%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	1.18185e-05	1.79353e-03	1.30230e-03	3.10764e-03	100.01%
Percentage	0.38%	57.71%	41.91%	100.00%	100.00%

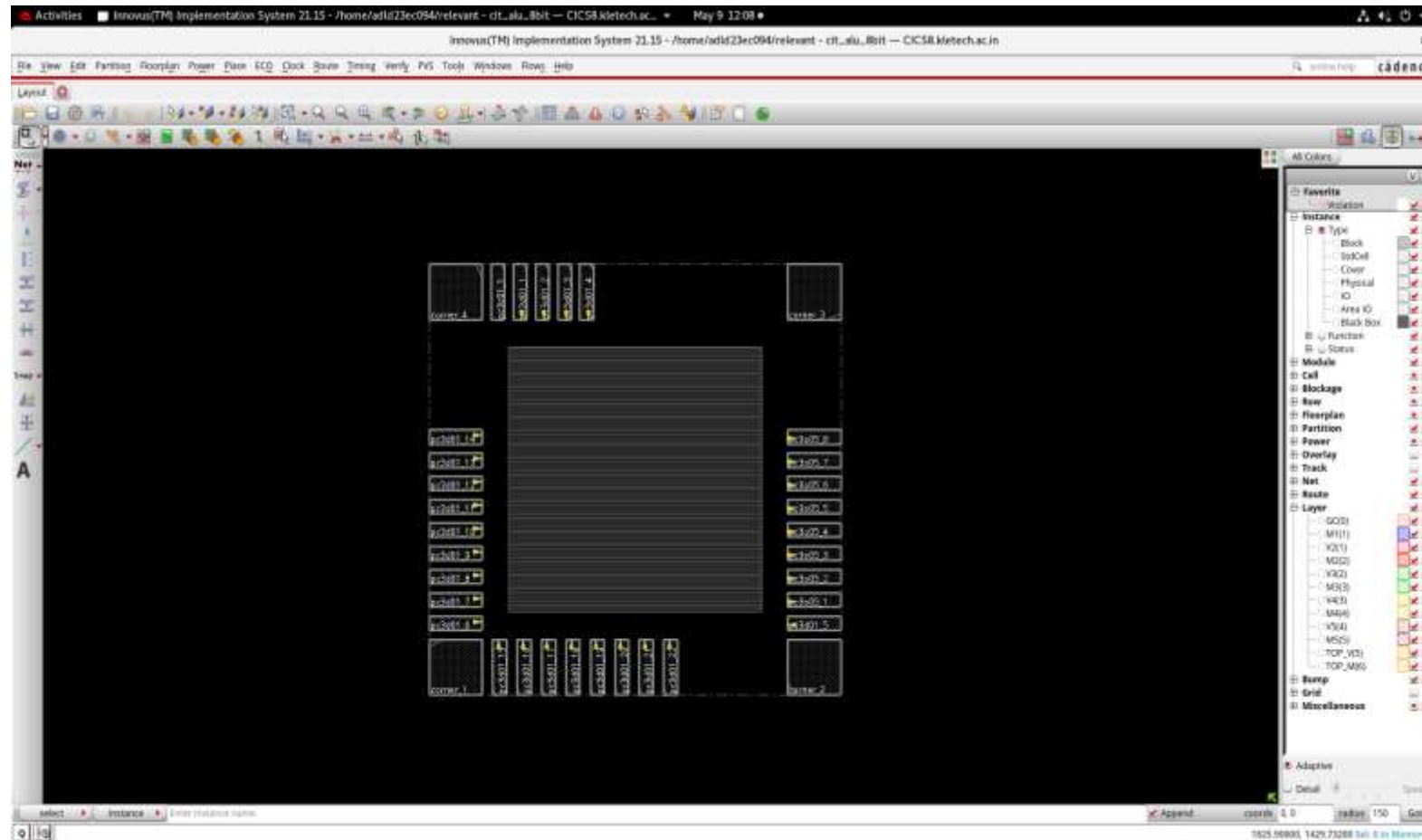
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LEC and GLS



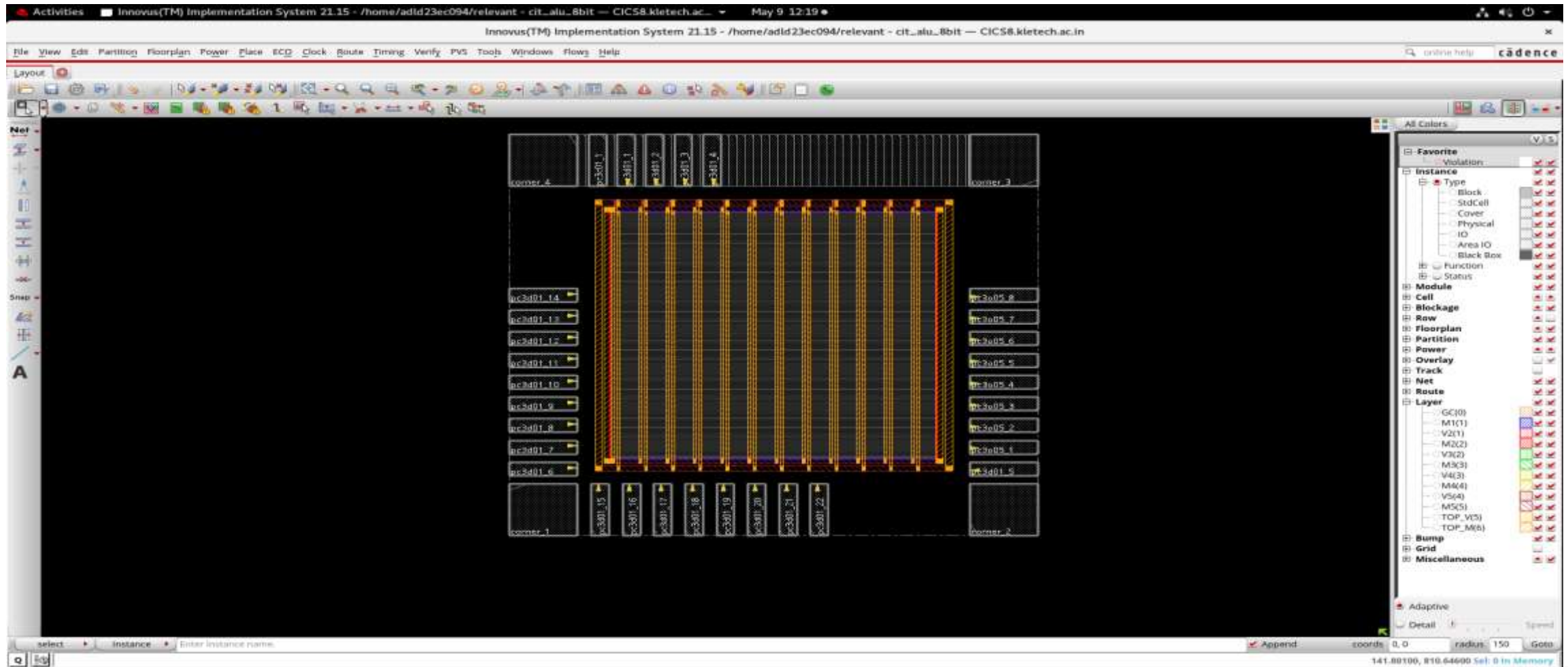
Proper GDS2 stream out with IO pad instantiation

Layout generation in Innovus



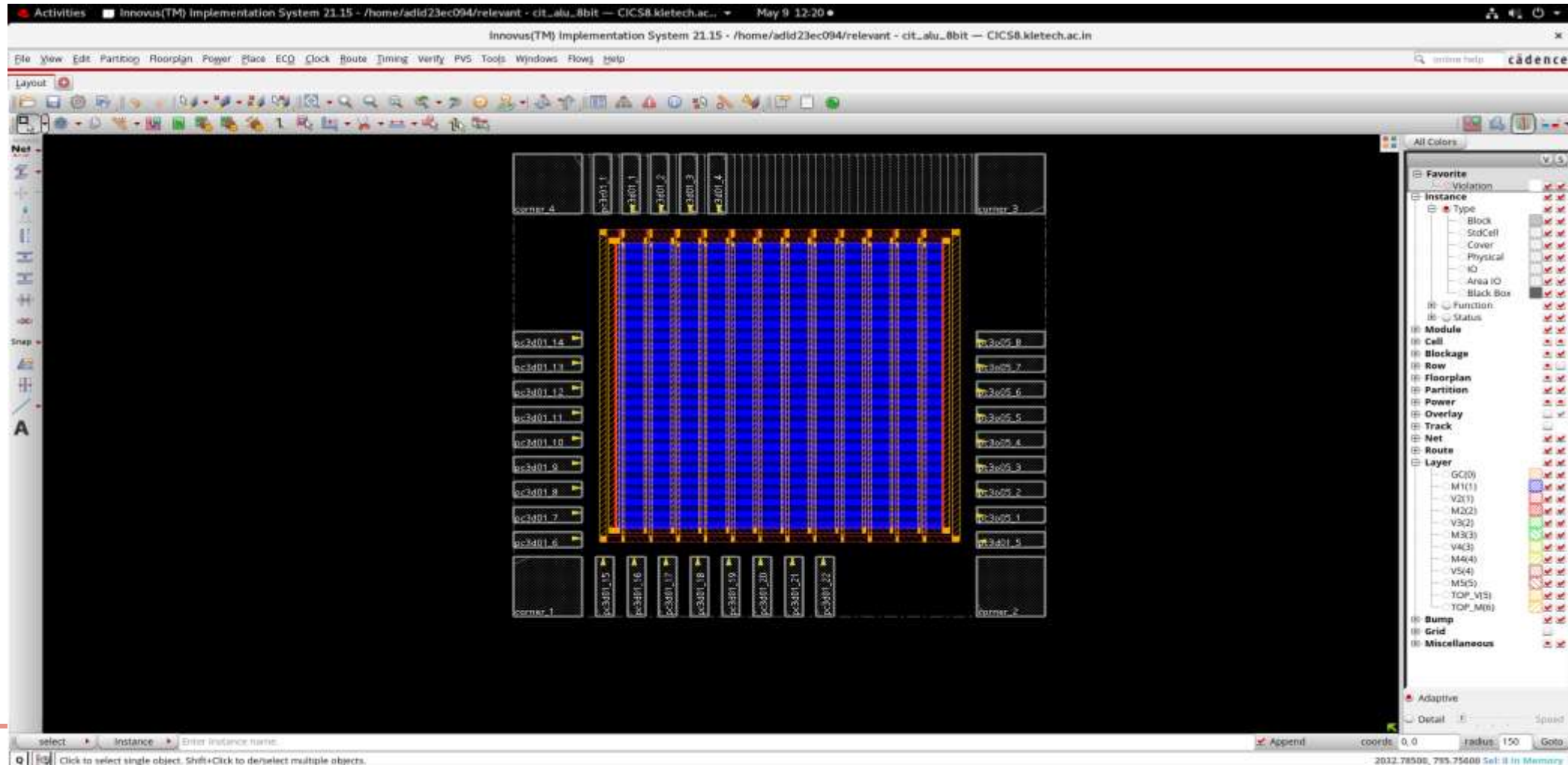
Proper GDS2 stream out with IO pad instantiation

Power Rail connections



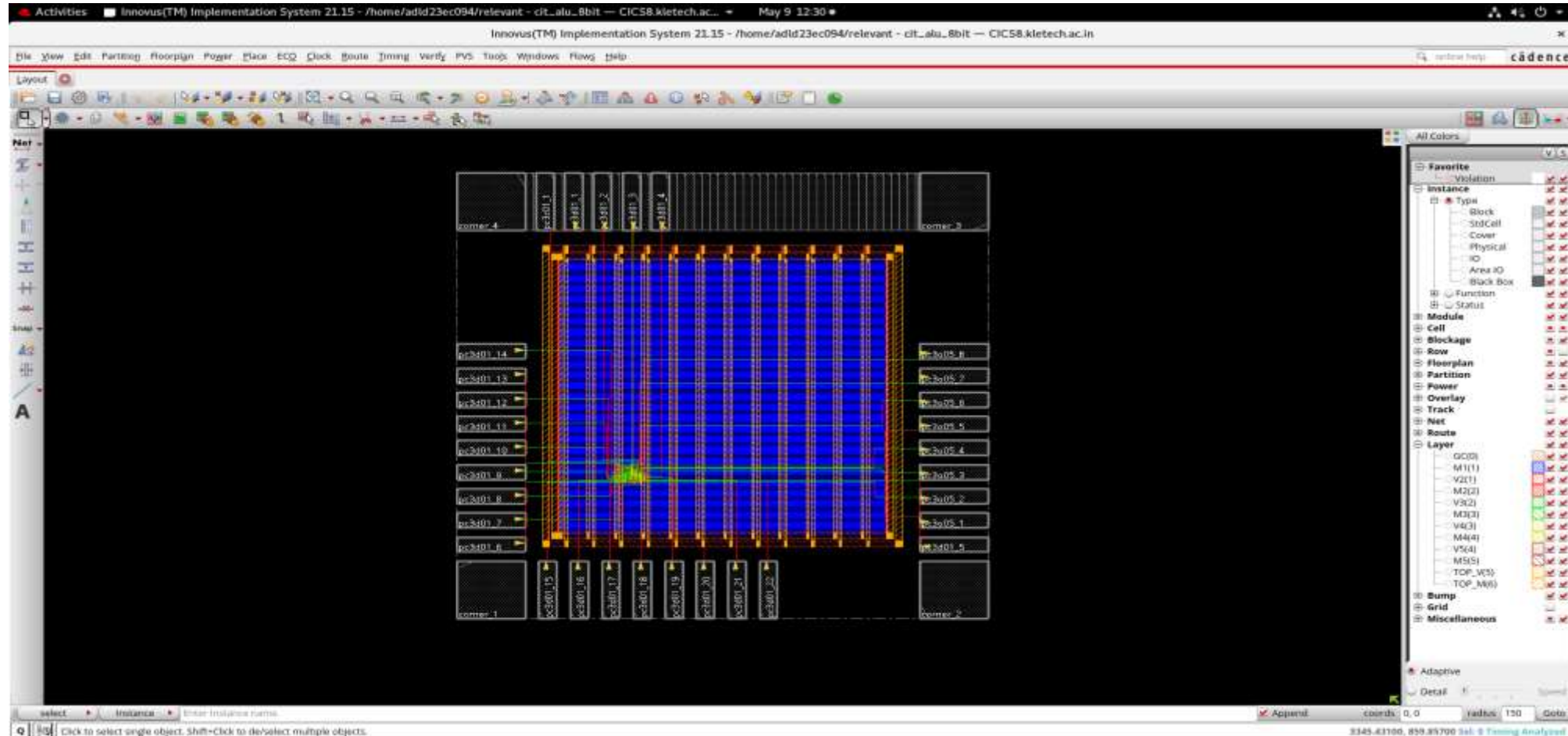
Proper GDS2 stream out with IO pad instantiation

After IO fillers



Proper GDS2 stream out with IO pad instantiation

Nano-Routing initiation (Pre-CTS)



Proper GDS2 stream out with IO pad instantiation

Post CTS



Proper GDS2 stream out with IO pad instantiation

GDS2 streamed in Virtuoso

